Xiaojun Wang

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Professional Summary

Staff GPU Engineer with 8+ years of industry experience developing GPU models, fixed-function graphics features, validation infrastructure, and GPU IP integrations. Combines a deep background in hardware simulation and graphics with active research/practice in NPU/AI accelerator architectures and AI Agent workflows.

Professional Experience

Staff GPU Engineer / Technical Lead

MediaTek — San Diego / San Jose, CA
May 2023 – Present

- Led **Arm Mali GPU IP integration** for SoC platforms, including IP-level validation, **graphics-driver** enablement, system bring-up, and cross-functional issue triage.
- Served as **Geometry High Level Model Lead** for an in-house GPU, driving geometry-pipeline modeling(Triangle assemble/setup, Clipping, Tiling, TBDR, DVS, Compression...) and integrating a texture-compression block.
- Worked as a **GPU Architect** on a confidential next-generation GPU program, defining and developing new architectural features.
- Performed **GPU benchmark**(Aztec, Manhattan...) validation and performance profiling using PVRTune to identify bottlenecks and support optimization. Designed and Implemented Top-level Vulkan tests.

Senior GPU Engineer

Qualcomm — San Diego / Santa Clara, CA
January 2018 – May 2023

- Conducted accurate high-level modeling and analysis for a large-scale C++ GPU simulator across Linux and Windows environments.
- Owned and trained engineers on two major fixed-function graphics blocks: **Tessellation** and **Primitive Control**.
- Developed features for the **CCU (Cache and Compression Unit)** and maintained source code across five fixed-function blocks in the graphics pipeline.
- Delivered two major features and more than 10 minor features spanning the 6th, 7th, and 8th generations of Adreno GPUs.
- Resolved more than 1,000 modeling bugs across multiple Qualcomm SoCs, improving simulation correctness and platform readiness.
- Partnered closely with RTL, design verification, and architecture teams to debug GPU issues and evaluate novel graphics features.
- Created and delivered more than 200 bit-accurate unit tests using DirectX 11, DirectX 12, and Vulkan, strengthening functional coverage for next-generation Qualcomm GPUs.

- Maintained a legacy sanity-regression test list containing more than 4,000 tests across multiple GPU performance tiers and release milestones.
- Developed and maintained an image-comparison tool used by more than 50% of relevant test cases, reducing test-development effort by approximately 15% and lowering manual-analysis errors.
- Expanded the image-comparison framework to support two additional image formats.
- Built more than 10 Python tools for image analysis, log analysis, regression triage, and workflow automation, improving engineering productivity and reducing human error.

Other Professional & Open-Source Projects

LuminError | Open-Source Real-Time Rendering & Game Engine
https://github.com/gpuwangge/LuminError | C++, Vulkan, Custom Ray Tracing (Oct 2025 – Present)

- Engine Architecture: Engineered a modular C++ Vulkan rendering engine with decoupled subsystems (core_renderer, core_resource, pipeline management) for hybrid graphics/compute workloads.
- Ray Tracing & AS Management: Implemented Vulkan KHR Ray Tracing, managing BLAS/TLAS builds, intersection shaders, and Next Event Estimation (NEE) sampling.
- Performance Optimization: Optimized descriptor sets, ray divergence, and memory bandwidth for complex scenes like Sponza.

Microsoft / Microsoft Research Asia — Software Engineering & Research Intern
Sunnyvale, CA | Beijing, China | 2008, 2009, 2013

- Built automated data-quality pipelines (SQL Server, C) for the Windows 8 News app and developed multithreaded HCI prototypes for Microsoft Surface.

Education

Ph.D., Computer Science (Computer Graphics) Michigan State University
East Lansing, MI, September 2009 – December 2017
Dissertation: *Fluid Animation on Deforming Surface Meshes*. Research in geometric processing, fluid simulation, deformable materials, and 3D mesh representation.
B.S., Automation Beihang University, Beijing, China, September 2004 – July 2008

Publications

Xiaojun Wang, Shiguang Liu, Yiying Tong. “Stain Formation on Deforming Inelastic Cloth.”
IEEE Transactions on Visualization and Computer Graphics.
DOI: 10.1109/TVCG.2017.2789203. Corpus ID: 51612072.
Ze Zhang, Xiaojun Wang, Yiying Tong. “Angle-Based Representation of Triangulated Surfaces.”
International Conference on Computer Graphics and Image Processing (CGIP), 2024.

Technical Skills

- Languages: C++/C, Python, SystemC, TLM, GLSL/HLSL
- Graphics & Compute: Vulkan, DirectX 11/12, OpenGL/ES, CUDA, OpenCL, Ray Tracing
- GPU Domains: Microarchitecture, Modeling, Driver Development, HW Verification, Parallel Computing, Performance Profiling (RenderDoc, PVRTune)
- AI & Infrastructure: AI Agents, AI Infrastructure, NPU Architecture Concepts